4.10 Properties of the Cumulative Distribution function

Recall that for the distribution function F of X, F(b) denotes the probability that the random variable X takes on a value that is less than or equal to b. The following are some properties of the cumulative distribution function (c.d.f.) F:

1. F is a nondecreasing function; that is, if $a<b$, then $F(a)\leq F(b)$.
2. $\lim_{b\to\infty} F(b)=1$.
3. $\lim_{b\to-\infty} F(b)=0$.
4. F is right continuous. That is, for any b and any decreasing sequence $b_n, n\geq 1$, that converges to b, $\lim_{n\to\infty} F(b_n)=F(b)$.

Property 1 follows, as was noted in **Section 4.1**, because, for $a<b$, the event $\{X\leq a\}$ is contained in the event $\{X\leq b\}$ and so cannot have a larger probability. Properties 2, 3, and 4 all follow from the continuity property of probabilities (**Section 2.6**). For instance, to prove property 2, we note that if b_n increases to ∞, then the events $\{X\leq b_n\}, n\geq 1$, are increasing events whose union is the event $\{X<\infty\}$. Hence, by the continuity property of probabilities,

$$\lim_{n\to\infty} P\{X\leq b_n\}=P\{X<\infty\}=1$$

which proves property 2.

The proof of property 3 is similar and is left as an exercise. To prove property 4, we note that if b_n decreases to b, then $\{X\leq b_n\}, n\geq 1$, are decreasing events whose intersection is $\{X\leq b\}$. The continuity property then yields

$$\lim_{n\to\infty} P\{X\leq b_n\}=P\{X\leq b\}$$

which verifies property 4.

All probability questions about X can be answered in terms of the c.d.f., F. For example,

$$P\{a<X\leq b\}=F(b)-F(a) \text{ for all } a<b \tag{10.1}$$

This equation can best be seen to hold if we write the event $\{X\leq b\}$ as the union of the mutually exclusive events $\{X\leq a\}$ and $\{a<X\leq b\}$. That is,

$$\{X\leq b\}=\{X\leq a\}\cup\{a<X\leq b\}$$

so

$$P\{X\leq b\}=P\{X\leq a\}+P\{a<X\leq b\}$$

which establishes **Equation (10.1)**.

If we want to compute the probability that X is strictly less than b, we can again apply the continuity property to obtain

$$P\{X<b\}=P(\lim_{n\to\infty}\{X\leq b-1n\})=\lim_{n\to\infty} P(X\leq b-1n)=\lim_{n\to\infty} F(b-1n)$$

Note that $P\{X<b\}$ does not necessarily equal F(b), since F(b) also includes the probability that X equals b.

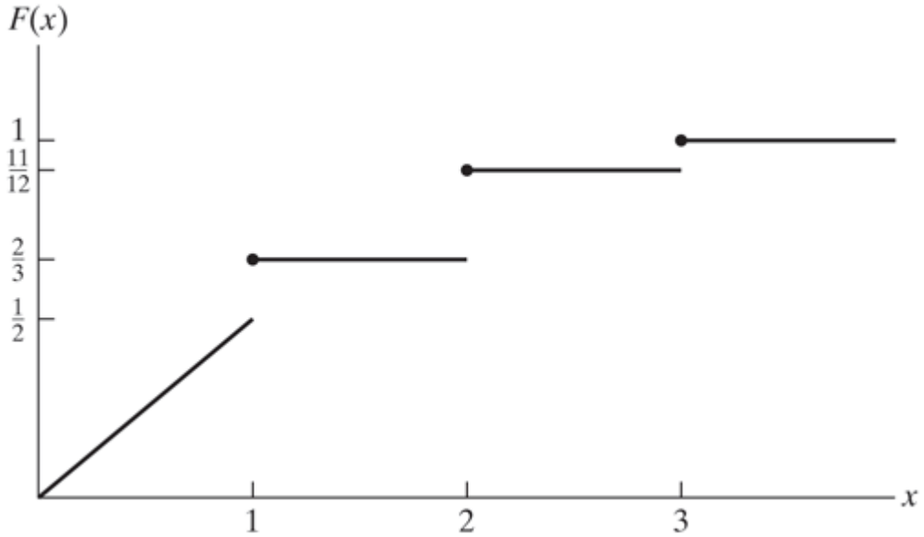
Example 10a

The distribution function of the random variable X is given by

F(x)={ 0x<0x20≤x<1231≤x<211122≤x<313≤x

A graph of F(x) is presented in **Figure 4.10**. Compute (a) P{X<3}, (b) P{X=1}, (c) P{X>12}, and (d) P{2<X≤4}.

Figure 4.10 Graph of f(x).



Solution

a. P{X<3}=limn P{X≤3−1n}=limn F(3−1n)=1112
b. P{ X=1 }=P{ X≤1 }−P{X<1}=F(1)−limn F(1−1n)=23−12=16

c. P{ X>12 }=1−P{ X≤12 }=1−F(12)=34

d. P{2<X≤4}=F(4)−F(2)=112

Summary

A real-valued function defined on the outcome of a probability experiment is called a *random variable*.

If X is a random variable, then the function F(x) defined by

$$F(x)=P\{X\leq x\}$$

is called the *distribution function* of X. All probabilities concerning X can be stated in terms of F.

A random variable whose set of possible values is either finite or countably infinite is called *discrete*. If X is a discrete random variable, then the function

$$p(x)=P\{X=x\}$$

is called the *probability mass function* of X. Also, the quantity E[X] defined by

$$E[X]=\sum x:p(x)>0xp(x)$$

is called the *expected value* of X. E[X] is also commonly called the *mean* or the *expectation* of X.

A useful identity states that for a function g,

$$E[g(X)]=\sum x:p(x)>0g(x)p(x)$$

The *variance* of a random variable X, denoted by Var(X), is defined by

$$Var(X)=E[(X-E[X])2]$$

The variance, which is equal to the expected square of the difference between X and its expected value, is a measure of the spread of the possible values of X. A useful identity is

$$Var(X)=E[X2]-(E[X])2$$

The quantity Var(X) is called the *standard deviation* of X.

We now note some common types of discrete random variables. The random variable X whose probability mass function is given by

$$p(i)=(ni)pi(1-p)n-ii=0,\ldots,n$$

is said to be a *binomial* random variable with parameters n and p. Such a random variable can be interpreted as being the number of successes that occur when n independent trials, each of which results in a success with probability p, are performed. Its mean and variance are given by

$$E[X]=npVar(X)=np(1-p)$$

The random variable X whose probability mass function is given by

$$p(i)=e-\lambda\lambda ii!i\geq 0$$

is said to be a *Poisson* random variable with parameter λ. If a large number of (approximately) independent trials are performed, each having a small probability of being successful, then the number of successful

trials that result will have a distribution that is approximately that of a Poisson random variable. The mean and variance of a Poisson random variable are both equal to its parameter λ. That is,

$$E[X]=Var(X)=\lambda$$

The random variable X whose probability mass function is given by

$$p(i)=p(1-p)i-1i=1,2,\ldots$$

is said to be a *geometric* random variable with parameter p. Such a random variable represents the trial number of the first success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X]=1pVar(X)=1-pp2$$

The random variable X whose probability mass function is given by

$$p(i)=(i-1r-1)pr(1-p)i-ri\geq r$$

is said to be a *negative binomial* random variable with parameters r and p. Such a random variable represents the trial number of the rth success when each trial is independently a success with probability p. Its mean and variance are given by

$$E[X]=rpVar(X)=r(1-p)p2$$

A *hypergeometric* random variable X with parameters n, N, and m represents the number of white balls selected when n balls are randomly chosen from an urn that contains N balls of which m are white. The probability mass function of this random variable is given by

$$p(i)=(mi)(N-mn-i)(Nn)i=0,\ldots,m$$

With p=m/N, its mean and variance are

$$E[X]=npVar(X)=N-nN-1np(1-p)$$

An important property of the expected value is that the expected value of a sum of random variables is equal to the sum of their expected values. That is,

$$E[\sum i=1nXi]=\sum i=1nE[Xi]$$

Problems

4.1. Two balls are chosen randomly from an urn containing 8 white, 4 black, and 2 orange balls. Suppose that we win $2 for each black ball selected and we lose $1 for each white ball selected. Let X denote our winnings. What are the possible values of X, and what are the probabilities associated with each value?
4.2. Two fair dice are rolled. Let X equal the product of the 2 dice. Compute P{X=i} for i=1,…,36.
4.3. Three dice are rolled. By assuming that each of the 63=216 possible outcomes is equally likely, find the probabilities attached to the possible values that X can take on, where X is the sum of the 3 dice.
4.4. Five men and 5 women are ranked according to their scores on an examination. Assume that no two scores are alike and all 10! possible rankings are equally likely. Let X denote the highest ranking achieved by a woman. (For instance, X=1 if the top-ranked person is female.) Find
P{X = i}, i=1, 2, 3,…, 8, 9, 10.
4.5. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?
4.6. In **Problem 4.5**, for n=3, if the coin is assumed fair, what are the probabilities associated with the values that X can take on?
4.7. Suppose that a die is rolled twice. What are the possible values that the following random variables can take on:

a. the maximum value to appear in the two rolls;
b. the minimum value to appear in the two rolls;
c. the sum of the two rolls;
d. the value of the first roll minus the value of the second roll?

4.8. If the die in **Problem 4.7** is assumed fair, calculate the probabilities associated with the random variables in parts (a) through (d).
4.9. Repeat **Example 1c** when the balls are selected with replacement.
4.10. Let X be the winnings of a gambler. Let p(i)=P(X=i) and suppose that
p(0)=1/3;p(1)=p(−1)=13/55;p(2)=p(−2)=1/11;p(3)=p(−3)=1/165

Compute the conditional probability that the gambler wins i, i=1,2,3, given that he wins a positive amount.
4.11. The random variable X is said to follow the distribution of Benford's Law if
P(X=i)=log10(i+1i),i=1,2,3,…,9

It has been shown to be a good fit for the distribution of the first digit of many real life data values.

a. Verify that the preceding is a probability mass function by showing that ∑i=19P(X=i)=1.
b. Find P(X≤j).

4.12. In the game of Two-Finger Morra, 2 players show 1 or 2 fingers and simultaneously guess the number of fingers their opponent will show. If only one of the players guesses correctly, he wins an amount (in dollars) equal to the sum of the fingers shown by him and his opponent. If both players guess correctly or if neither guesses correctly, then no money is exchanged. Consider a specified player, and denote by X the amount of money he wins in a single game of Two-Finger Morra.

a. If each player acts independently of the other, and if each player makes his choice of the number of fingers he will hold up and the number he will guess that his opponent will hold up in such a way that each of the 4 possibilities is equally likely, what are the possible values of X and what are their associated probabilities?
b. Suppose that each player acts independently of the other. If each player decides to hold up the same number of fingers that he guesses his opponent will hold up, and if each player is equally likely to hold up 1 or 2 fingers, what are the possible values of X and their associated

probabilities?

4.13. A salesman has scheduled two appointments to sell vacuum cleaners. His first appointment will lead to a sale with probability .3, and his second will lead independently to a sale with probability .6. Any sale made is equally likely to be either for the deluxe model, which costs $1000, or the standard model, which costs $500. Determine the probability mass function of X, the total dollar value of all sales.

4.14. Five distinct numbers are randomly distributed to players numbered 1 through 5. Whenever two players compare their numbers, the one with the higher one is declared the winner. Initially, players 1 and 2 compare their numbers; the winner then compares her number with that of player 3, and so on. Let X denote the number of times player 1 is a winner. Find $P\{X=i\}, i=0,1,2,3,4$.

4.15. The National Basketball Association (NBA) draft lottery involves the 11 teams that had the worst won–lost records during the year. A total of 66 balls are placed in an urn. Each of these balls is inscribed with the name of a team: Eleven have the name of the team with the worst record, 10 have the name of the team with the second-worst record, 9 have the name of the team with the third-worst record, and so on (with 1 ball having the name of the team with the 11th-worst record). A ball is then chosen at random, and the team whose name is on the ball is given the first pick in the draft of players about to enter the league. Another ball is then chosen, and if it "belongs" to a team different from the one that received the first draft pick, then the team to which it belongs receives the second draft pick. (If the ball belongs to the team receiving the first pick, then it is discarded and another one is chosen; this continues until the ball of another team is chosen.) Finally, another ball is chosen, and the team named on the ball (provided that it is different from the previous two teams) receives the third draft pick. The remaining draft picks 4 through 11 are then awarded to the 8 teams that did not "win the lottery," in inverse order of their won–lost records. For instance, if the team with the worst record did not receive any of the 3 lottery picks, then that team would receive the fourth draft pick. Let X denote the draft pick of the team with the worst record. Find the probability mass function of X.

4.16. A deck of n cards numbered 1 through n are to be turned over one a time. Before each card is shown you are to guess which card it will be. After making your guess, you are told whether or not your guess is correct but not which card was turned over. It turns out that the strategy that maximizes the expected number of correct guesses fixes a permutation of the n cards, say 1,2,…,n, and then continually guesses 1 until it is correct, then continually guesses 2 until either it is correct or all cards have been turned over, and then continually guesses 3, and so on. Let G denote the number of correct guesses yielded by this strategy. Determine $P(G=k)$.

Hint: In order for G to be at least k what must be the order of cards 1,…,k.

4.17. Suppose that the distribution function of X is given by

F(b)={ 0b<0b40≤b<112+b−141≤b<211122≤b<313≤b

a. Find $P\{X=i\}, i=1,2,3$.
b. Find $P\{12<X<32\}$.

4.18. Four independent flips of a fair coin are made. Let X denote the number of heads obtained. Plot the probability mass function of the random variable X−2.

4.19. If the distribution function of X is given by

F(b)={ 0b<0120≤b<1351≤b<2452≤b<39103≤b<3.51b≥3.5

calculate the probability mass function of X.

4.20. A gambling book recommends the following "winning strategy" for the game of roulette: Bet $1 on red. If red appears (which has probability 1838), then take the $1 profit and quit. If red does not appear and you lose this bet (which has probability 2038 of occurring), make additional $1 bets on red on each of the next two spins of the roulette wheel and then quit. Let X denote your winnings when you quit.

a. Find $P\{X>0\}$.
b. Are you convinced that the strategy is indeed a "winning" strategy? Explain your answer!
c. Find $E[X]$.